

Beyond Risk: Firm Financing and Interest Rates*

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Abstract

Firm financing is a central component of modern macroeconomic theory, yet the empirical determinants of interest rates are not well understood. We study these using a novel loan-level dataset. Interest rates vary substantially across firms, even on observationally equivalent loans. Default risk—the core source of dispersion in standard models—can only account for 15% of variation. Bank heterogeneity and relationship banking have similarly limited explanatory power. Instead, most dispersion reflects persistent idiosyncratic firm-level heterogeneity—accounting for 27% of variation—that cannot be explained by observable firm characteristics. Our findings suggest that there are substantial frictions in credit markets outside existing frameworks.

Keywords: Firm financing, interest rates, financial frictions.

JEL Codes: E43; E44; G32.

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1 Introduction

Firm financing is a central component of modern macroeconomic theory. It is a key ingredient in theories of business cycles, monetary policy transmission, economic development, misallocation, firm dynamics, inequality, tax policy and macroeconomic aspects of international trade, among other things.¹ There is broad agreement that frictions in firm financing shape the allocation of capital in the economy and are quantitatively relevant for many macroeconomic outcomes. In these theories, interest rates are a key price. They determine firms borrowing decisions, affecting investment choices, firm-level outcomes and the aggregate economy. Despite their importance, the empirical determinants of interest rates are still not well understood.

This paper seeks to fill this gap and understand the sources of interest rate dispersion across firms in the context of bank loans. Our first contribution is to quantify how much dispersion can be accounted for by the leading mechanisms in existing frameworks. We begin with the dominant risk-based view of firm financing and show that default risk can only account for 15% of the observed variation in borrowing costs. We then turn to other prominent channels, namely heterogeneity across banks and firm-bank relationships. Both matter, but each accounts for only about 10% of overall dispersion. This leaves the majority of the variation originating from sources outside of the leading frameworks.

Our second contribution is to show what accounts for the remaining variation. We document that the largest source of interest rate dispersion, accounting for over one-quarter of the total variance, is persistent idiosyncratic firm-level heterogeneity that is unrelated to observable firm characteristics. The persistent differences in rates across firms are not absorbed by rich controls for firm size, age, industry, location, financial position, credit history, productivity, production input mix, and many other observables,

¹Some examples are as follows. Business cycles: Cooley et al. (2004), Jermann & Quadrini (2012), Khan & Thomas (2013), Gilchrist et al. (2014), Buera & Moll (2015) and Arellano et al. (2019). Monetary policy transmission: Kiyotaki & Moore (1997), Bernanke & Gertler (1989), Bernanke et al. (1999) and Ottonello & Winberry (2020). Economic development: Buera et al. (2011), Moll (2014) and Itskhoki & Moll (2019). Misallocation: Midrigan & Xu (2014) and Gopinath et al. (2017). Firm dynamics: Cooley & Quadrini (2001), Albuquerque & Hopenhayn (2004) and Arellano et al. (2012). Inequality and tax policy: Boar & Midrigan (2023) and Guvenen et al. (2023). International trade: Antras & Caballero (2009), Manova (2013), Leibovici (2021) and Spencer (2022).

and they persist even when firms obtain new loans and switch banks. Our third contribution is to use this evidence to narrow the set of plausible explanations for that latent, persistent firm-level heterogeneity. We find little support for explanations based on local bank market power, the joint pricing of multiple loans to the same firm, interactions between entrepreneurs' personal and business finances, or unobservable demand shocks. Instead, the evidence points to search-related frictions as the most plausible source of the variation across firms.

The key to our analysis is a detailed new credit registry which we merge with comprehensive firm balance sheet and income statement data, additional risk measurement from the banking regulator, and geographic information on the location of bank branches. There are a number of important features of the data that make the analysis possible. It provides detailed measures of default risk that match up with the theoretical moments needed for pricing. It contains over half a million loans, with many observations at the bank level, the firm level, and even within firm-bank relationships. This facilitates the estimation of bank, firm and relationship effects on rates. Finally, with detailed information on firm characteristics, we can evaluate the factors behind differences in rates at the firm level.

The analysis starts by taking the core theories for frictional firm financing from the macroeconomic literature as a framework. While these theories are not designed with the full distribution of rates in mind, we are interested in quantifying how far the data departs from them. Do the theories merely omit second-order factors and, if not, what are the main sources of rate variation that are ignored? These core theories are based on the classic view of interest rates, formalized by [Merton \(1974\)](#), which looks at interest rates from the perspective of default risk. Most models either have lending without risk, so that all firms pay the same risk free rate (e.g. [Buera et al., 2011](#); [Moll, 2014](#)),² or there is default risk and loans are priced so that lenders earn the risk free rate in expectation (e.g. [Bernanke et al., 1999](#)). Under the second approach, interest rates differ across firms

²Various modeling techniques are used to achieve this. The most common approach is to impose a collateral constraint, assuming that a firm's existing assets constrain how much it can borrow, and that either it cannot default, or the environment is such that it will not choose to default in equilibrium (e.g. [Moll, 2014](#)). A related approach assumes that loans are uncollateralized, but that there are endogenous borrowing limits so that firms never default (e.g. [Buera et al., 2011](#)).

depending on their risk.

The first stage of the empirical analysis shows that the data departs substantially from this risk-based perspective. Risk free lending does not fit the data well because there is a lot of heterogeneity in interest rates after controlling for a rich array of loan characteristics. This is true for both collateralized and uncollateralized loans. To quantify the importance of default risk, we take the relevant pricing equation directly to the data using our empirical measures of default probabilities and loss given default. These characteristics of loans can only account for 15% of the variation in interest rates that we observe. Overall, once we control for loan characteristics and default risk, approximately 65% of the variation in interest rates remains unaccounted for. This corresponds to an unexplained standard deviation of rates of 1.46 percentage points.

The second stage of the analysis investigates potential explanations for the additional dispersion in interest rates. We first consider the role of banks, as they have received the most attention as an additional source of rate variation in the literature. This research has studied how bank characteristics like market power, cost heterogeneity and portfolio composition can affect loan prices (e.g. Klein, 1971; Monti, 1972; Ho & Saunders, 1981; Diamond, 1984), and there is a long history of research on the effect of firm-bank relationships on credit conditions (e.g. Sharpe, 1990; Hoshi et al., 1991; Petersen & Rajan, 1995).³ These ideas have been incorporated into some macroeconomic models.⁴ We evaluate the quantitative relevance of these theories for the total variation in interest rates. They are quantitatively relevant, each accounting for approximately 10% of the variation in rates. However, nearly half of the variation remains unexplained, implying that there are substantial frictions in financial markets outside of the leading frameworks.

We show that over 30 percentage points of this residual variation can be attributed to persistent idiosyncratic firm-level heterogeneity above and beyond default risk, and that most of this (27 percentage points) is due to latent firm fixed effects. These latent fixed effects are the largest source of rate variation in the data. This result implies that if two

³For additional empirical evidence of these effects see, for example, Petersen & Rajan (1994), Berger & Udell (1994), Berger & Udell (1997), Berger & Udell (1998), Wheelock & Wilson (2012), Wheelock & Wilson (2018), Darmouni (2020) and Benetton (2021).

⁴For examples of macroeconomic models with bank heterogeneity and relationship banking, see Corbae & D’Erasmus (2021) and Dempsey & Faria-e Castro (2025), respectively.

observationally equivalent firms approach the same bank for the same loan, they obtain different rates, and these differences persist across subsequent loans, even when firms switch banks. We include rich controls for a wide variety of time-varying and immutable firm characteristics, so these effects are not due to heterogeneity in firm age, size, industry, location, financial position, credit history, productivity, production input mix and many other observables.

We consider a range of potential explanations for the large latent firm fixed effects in interest rates. The empirical results provide clear requirements: (1) the explanation must be based on an idiosyncratic, firm-specific source; (2) it must be able to generate highly persistent rate differences across firms; and (3) it must have a low correlation with the broad range of observable firm characteristics that we control for. There are a number of potential explanations based on bank behavior that relate to market power, geographic differences in markets, pricing multiple loans to a firm collectively rather than individually, or interactions between the pricing of loans to firms and personal loans to their owners or managers. We provide evidence against these channels. Variation in demand for loans due to unobservable shocks to investment opportunities, costs or liquidity is another possibility, but we also provide evidence against this. Instead, we argue that the market for bank loans to firms is an institutional setting that naturally gives rise to search frictions and that this type of theory has the necessary ingredients to fit the data well. Within a search setting, rate variation could come from differences in latent search ability, effects at the loan officer level, or differentiation based on non-economic factors such as socio-demographic characteristics or personal relationships.

Disentangling these search-related sources of rate variation is an important question for future research. Differences in search ability would typically be considered a technological feature of the economy, while effects at the loan officer level and differentiation on non-economic factors are more likely to be wedges generating misallocation. This is particularly relevant since we document persistent differences in rates that typically have a larger impact on investment than transitory differences.

Contribution to the literature There is a long history of theoretical work on the determinants of interest rates and their heterogeneity across firms. Leading strands of this

research have focused on the relevance of default risk, bank heterogeneity, and relationship banking.⁵ In macroeconomic models, default risk has been the most common source of rate heterogeneity (e.g. [Bernanke et al., 1999](#)), however there is also work incorporating bank effects and relationship banking (e.g. [Corbae & D’Erasmus, 2021](#); [Dempsey & Faria-e Castro, 2025](#)). Our contribution is to quantify how much variation in interest rates can be attributed to each of these channels, show that a large share of variation comes from elsewhere, and provide evidence on the source of this variation.

There is, of course, a large existing empirical literature studying the determinants of interest rates. This literature has, in particular, focused on how rates vary as a result of default risk (e.g. [Durrani et al., 2022](#)), bank heterogeneity (e.g. [Santos, 2011](#)) and relationship banking (e.g. [Petersen & Rajan, 1994](#)). Compared to this literature, we focus on a different question. Rather than evaluating how a particular factor generates variation in rates, we quantify the overall contribution of each channel and provide guidance on where the substantial additional variation in rates comes from.

The most closely related paper is [Amiti et al. \(2026\)](#), which also studies the overall variation in interest rates on bank loans to firms. Both papers show that substantial dispersion in borrowing costs remains after accounting for loan characteristics, default risk, bank characteristics, and relationship variables. Our paper additionally quantifies how much of overall interest rate dispersion can be attributed to each of these sources, and uses a theoretically-based approach to measuring the role of default risk. The papers also differ in how they analyze the residual variation in rates. [Amiti et al. \(2026\)](#) interpret it through the lens of the [Burdett & Judd \(1983\)](#) model, structurally estimating the search costs required to match the data, and providing empirical evidence to support this interpretation. We instead show that the dominant source of the remaining dispersion is persistent idiosyncratic firm-level heterogeneity that is uncorrelated with observables, rule out several prominent potential explanations for what drives this heterogeneity, and argue that search-related frictions are the most plausible interpretation.

[Faria-e-Castro et al. \(2024a\)](#) studies rate variation on bank loans in the US. The analysis

⁵For example, see [Merton \(1974\)](#) on default risk; [Klein \(1971\)](#) [Monti \(1972\)](#), [Ho & Saunders \(1981\)](#) and [Diamond \(1984\)](#) on bank heterogeneity; and [Sharpe \(1990\)](#), [Hoshi et al. \(1991\)](#), and [Petersen & Rajan \(1995\)](#) on relationship banking.

differs from ours by focusing on comparing the cost of bank loans to corporate bonds, and within-firm variation in rates. [Cavalcanti et al. \(2023\)](#) also studies rate variation, but in the context of a developing country (Brazil) in which interest rates look very different from those in our setting.⁶ That paper also differs by focusing on how rates vary with observable firm characteristics, which is not a quantitatively important feature of our data. Consistent with our interpretation of the source of rate variation, [Cerqueiro et al. \(2011\)](#) and [Mazet-Sonilhac \(2025\)](#) also argue that discretion in loan pricing and search frictions, respectively, are important sources of interest rate variation. While the message of [Cerqueiro et al. \(2011\)](#) is consistent with our conclusions, it does not have sufficiently detailed data to study and quantify the breadth of channels that we do. [Mazet-Sonilhac \(2025\)](#) is focused specifically on documenting the existence of search frictions when firms lend from banks, rather than evaluating overall rate variation.

The remainder of the paper is structured as follows. Section 2 describes the data. Section 3 evaluates how much variation in interest rates can be accounted for by loan characteristics and default risk. Section 4 explores the importance of banks and firm-bank relationships for dispersion in interest rates. Section 5 evaluates the relevance of firm characteristics for interest rate differences. Section 6 investigates the source of fixed latent variation in interest rates across firms. Section 7 concludes.

2 Data

The analysis uses two primary datasets, a credit registry covering the universe of credit instruments issued by banks to firms in Portugal, and a dataset with balance sheet and income statement information for the universe of incorporated firms.⁷

The credit registry has a monthly frequency, covering January 2019 to December 2022. For each month it provides detailed information about the characteristics of all outstanding credit instruments issued by banks to firms.⁸ The set of credit instruments is broad,

⁶Interest rates are approximately an order of magnitude higher in the Brazilian data, and their standard deviation is nearly two orders of magnitude larger.

⁷The names of the datasets are *Central de Responsabilidades de Crédito do Banco de Portugal* and *Informação Empresarial Simplificada*.

⁸There is a small number of loans to public sector firms in the data. We drop these, restricting the sample to loans to private sector firms.

encompassing all types of contracts through which firms borrow from banks including traditional business loans, a range of credit lines, credit cards, mortgages and auto loans. The data has information about the current characteristics of each instrument including the amount outstanding, the interest rate, the duration of the instrument, whether the interest rate is fixed or variable and the time horizon for changes to it, and information about any collateral, among other things. Importantly, the data also includes estimates for the probability of default of firms and loss given default for loans. These are provided by the banks issuing credit instruments and will be a central component of our analysis. After an instrument is initiated, the registry records information about the repayments every month, and whether the instrument is in arrears or default. For credit lines there is information about their total size and the amount that has been drawn. The information about credit instruments extends to their characteristics at origination and the time (month) of origination. This backward-looking feature means that despite the dataset starting in 2019, it has considerable information about loans issued before this date.

The other dataset provides balance sheet and income statement information for the universe of incorporated firms at an annual frequency, excluding financial corporations and entities classified in general government. Incorporated firms account for 33% of firms and 72% of employment in the economy.⁹ The data comes from mandatory reporting of financial statements to the Ministry of Finance each year. It covers 2010 to 2022, so it overlaps with nearly all instruments in the credit registry going back to their origination dates.¹⁰

To construct our dataset we merge the credit registry with the firm data, dropping instruments that cannot be matched to the firm data in order to ensure a consistent sample across our analysis.¹¹ Only 32% of total credit in the data is excluded from the matched sample as a result of this, since the firm data only omits very small firms. The main types of instruments are traditional bank loans (35% of total credit), factoring and confirming (49%), credit cards (7%), auto loans (5%) and mortgages (2%). For our primary analysis,

⁹Only sole proprietorships and non-employer firms without a separate legal status from their owner are excluded. See the Appendix for details of the coverage of our sample.

¹⁰The share of loans in the credit registry with an origination date prior to 2010 is 2.3%.

¹¹Instruments are dropped because either they were initiated before 2010, when the firm data starts, or were issued to unincorporated firms outside the firm data.

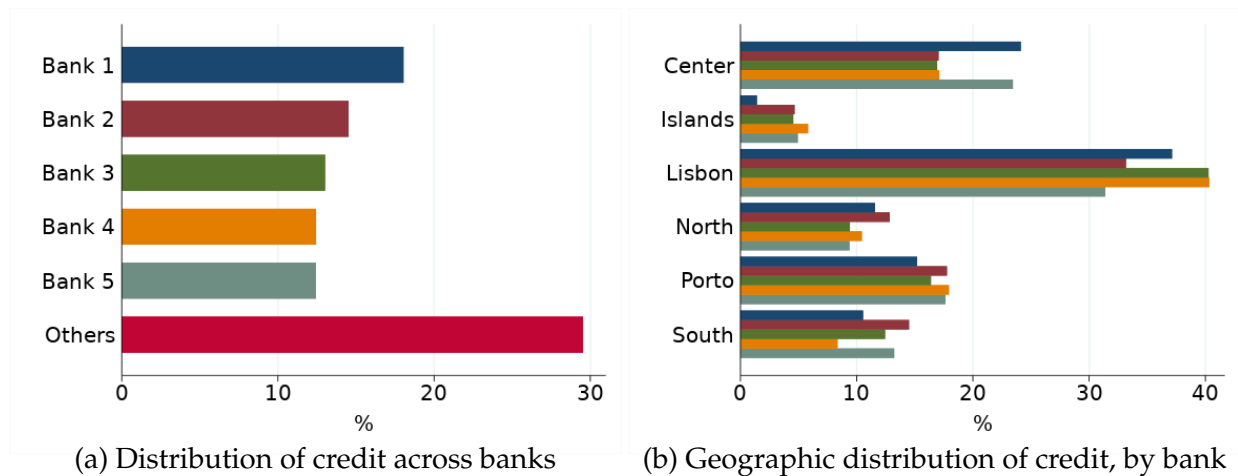


Figure 1: Distribution of firm financing credit across banks. This figure presents moments of the distribution of firm financing loans by value in December 2019. Panel (a) shows the share of credit accounted for by each bank. Banks other than the largest five are grouped together. Panel (b) shows the distribution of credit across regions for each of the five main banks. The color for each bank is the same as in panel (a).

we restrict the sample to traditional bank loans, as these are analogous to firm financing in models and this ensures a more homogeneous sample.¹² We will refer to these as “firm financing loans” or simply “loans” from now on, and all analysis will be for this sample, unless stated otherwise. Nevertheless, we show in the Appendix that our main results are robust to using all credit instruments.

The market for firm financing loans consists of five main banks, with each accounting for 12% to 18% of the total value of loans in our sample (Figure 1(a)). There are 157 smaller banks that collectively provide the remaining 29%.¹³ The five main banks are the main providers of both firm and consumer credit. They are large, national, general-purpose banks with similar operations to each other. To illustrate this, panel (b) shows that their loans to firms have a similar geographic distribution, and in the Appendix we confirm that the distributions are similar across industry and firm size as well.

Our sample has 150,700 firms. Relative to the population of firms in the industries covered by the data, the sample is skewed towards larger firms. Nearly all non-employer

¹²We are excluding the small share of renegotiated and renewed loans in the sample to avoid history dependence in interest rates.

¹³The market shares of the largest banks in other Euro Area 19 countries are similar. In the Appendix we show that share of banking sector assets held by the largest three and five banks in Portugal was similar to the average for the Euro Area 19 countries from 2010 to 2021.

firms are excluded because they are not incorporated and therefore not covered by the firm data. Larger firms are also more likely to have bank credit. Overall, our sample covers 37% of employer firms, 52% of total employment and 55% of aggregate output in 2019.¹⁴ The average number of employees of our firms is 11.3, compared to 9.4 for all employer firms.

In terms of overall credit for firms, bank financing is the most important source. From 2010 to 2022 it represented 86% of total credit to non-financial corporations in Portugal. This degree of dependence on banks is typical across Europe. Over the same period, the average share of credit to non-financial corporations that came from banks in the Euro Area 19 countries was 89%.¹⁵ While a substantial amount of firm financing comes from public debt markets in the U.S., these play a much smaller role in Europe.

To further illustrate the characteristics of the firms in the sample we present summary statistics in Table 1. The statistics are within firm averages over the years that each firm is in the sample. The balance sheet and income statement variables reflect the characteristics of the population of firms. For the credit registry variables, the median firm has €119,830 in loans. The median number of loans is four and the median share of a firm's loan value that is collateralized is 78%. The majority of firms have loans from only one bank, but many firms have more bank relationships. At the 90th percentile a firm averages 3 banking relationships over the years that it is in the sample.

Figure 2 presents summary statistics for all loans in the sample at their initiation date. Throughout the paper, we focus on evaluating the determinants of interest rates at loan initiation.¹⁶ There are 598,028 loans in total. The bulk of these (78.5%) were originated from 2018 onwards, with 19.2% originated from 2014–17, and the rest distributed over earlier years.¹⁷ The average interest rate is 3.01% with a standard deviation of 2.25%.¹⁸ The median loan is for €44,751. The right tail of the distribution is for much larger loans,

¹⁴Additional details of the comparison of our sample to the population of firms are provided in the Appendix. We also benchmark our sample against the size distribution of employer firms in the US.

¹⁵See the Appendix for additional details.

¹⁶For some loans, the rate varies over time. We do not use this variation because it results in a less homogeneous sample that raises additional empirical complications.

¹⁷Recall that the data starts in September 2018, with loans originated earlier entering the sample if they were still active at this date. This is why origination dates are skewed towards recent years.

¹⁸Throughout the paper, interest rates are always stated on a per annum basis.

	Average	p25	p50	p75	p90	St. dev.	# firms
Value of loans (€'000s)	1,063.31	38.24	119.83	378.79	1387.72	(9848.76)	150,700
# of loans	7.07	3.00	4.00	8.00	14.00	(13.29)	
Av. loan value (€'000s)	123.73	11.35	24.93	58.46	183.84	(908.95)	
% collateralized	67	40	78	100	100	(35)	
# of banks	1.63	1.00	1.00	2.00	3.00	(1.14)	
Employees	12.53	1.50	3.75	8.75	21.00	(123.86)	
Total assets (€'000s)	2,277.66	101.10	261.73	778.33	2478.56	(65527.84)	
Revenue (€'000s)	1,657.86	87.60	228.28	649.41	1931.82	(40855.78)	
Leverage (%)	103	49	71	91	125	(255)	
Age	15.15	5.50	11.50	21.00	31.50	(13.10)	

Table 1: Descriptive statistics for firms. This table provides statistics for firms in the sample. For each variable, the average, four percentiles and the standard deviation are given. All moments are at the firm level and are averaged within firm over the years that each firm is in the sample. So, for example, the value of loans is the total value of a firm's loans, % collateralized is the share of a firm's loans that are collateralized, and # of banks is the number of banks that a firm has loans from. Value of loans, total assets and revenue are in thousands of 2016 Euros.

with the 90th percentile having a value of €338,872. The maturities of loans range one month to 30 years. Sixty-eight percent of loans have a maturity of less than 5 years and they account for 48% of total loan value. For collateral, the distribution is binary. Forty eight percent of loans are fully collateralized and virtually all other loans are uncollateralized. The most common types of collateral are guarantees and financial assets (51.7% and 26.4% by loan value, respectively), with real estate and other collateral also being used. Throughout the paper the analysis focuses on interest rates rather than spreads, because spreads are not always reported, and the data shows that there is almost no variation in the implied financing costs of banks, conditional on maturity and loan origination date.¹⁹ Since we control for time of origination and maturity in our analysis, using spreads instead of interest rates does not affect the results.

In addition to the main data, we use two supplementary datasets. As an alternative measure of the probability of default for firms, we use an annual dataset from the banking regulator with probability of default estimates for all firms in the sample for 2010–2022.²⁰

¹⁹See the Appendix for details on this.

²⁰The methodology for estimating these probabilities of default is explained in Antunes et al. (2016). The dataset is called SIAC (*Sistema Interno de Avaliação de Crédito*).

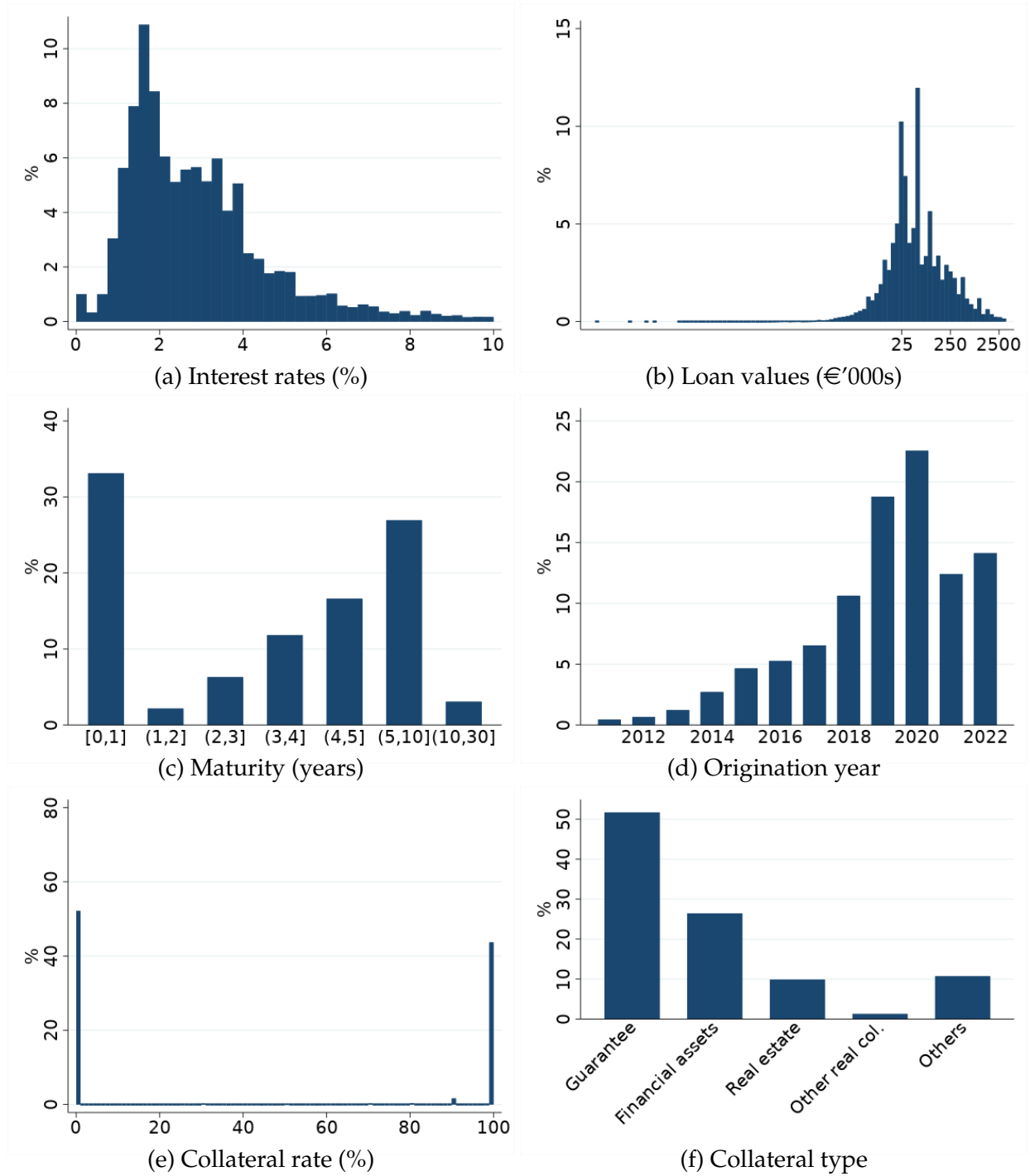


Figure 2: Distributions of loan characteristics at origination. Panels (a), (b), (c), (d), (e) and (f) are the distributions of interest rates, loan values (in thousands of Euros), maturity (in years), origination years, collateral rates and collateral types, respectively. The scale for loan values in panel (b) is in log units. The distribution in panel (f) is conditional on loans having some collateral.

We also use data on the postcodes of all bank branches for the same years.²¹ When

²¹This data is a supplement to the balance sheet and income statement data from Banco de Portugal.

merged with the location of each firm, this data provides information on the banking market around each firm. Note that despite having the location of all branches, the credit registry does not record which branch each loan is from, only the bank. The postcodes are very granular, with seven digits. They correspond to an average of one postcode per 0.70 square kilometers for the country, and one per 0.09 square kilometers for the Lisbon metropolitan area (0.27 and 0.03 square miles, respectively).

3 Risk free lending and default risk

The vast majority of macroeconomic models in which firms face external financing frictions think about interest rates from the perspective of risk. Either there is no risk, and all firms pay a common risk free rate, or there is heterogeneous default risk, and firms pay different rates to compensate banks for this. In this section we assess whether this approach provides a good first-order approximation to the data, or not.

3.1 Risk free lending

The first step is to assess how far the data is from risk free lending, as this is the simplest case and the most common assumption in macroeconomic models. In these models, risk free lending is typically achieved through one of two approaches. The most popular is to assume that there is a collateral constraint that limits how much a firm can borrow as a function of its assets. This takes the form

$$b_{i,t+1} \leq \lambda_{i,t} a_{i,t}$$

where $b_{i,t+1}$ is loan size of firm i from period t to $t + 1$, $a_{i,t}$ is the period t assets of the firm and $\lambda_{i,t} > 0$ determines the tightness of the constraint. Lending is risk free either because the collateral can be claimed in the case of default, or because default is ruled out by assumption. This approach dates back to [Kiyotaki & Moore \(1997\)](#) and is particularly common in research studying business cycles, firm dynamics and capital misallocation. Under the second approach, it is assumed that there is an endogenous borrowing limit, with banks not lending more to a firm than it will be able to repay with certainty (e.g.

Buera et al., 2011). Both approaches result in no default and firms paying a common risk free rate.

The data on collateralization rates aligns cleanly with these theories. Figure 2(e) in Section 2 shows that almost 50% of loans are fully collateralized, supporting the idea of no risk under the first theory. Nearly all of the remaining loans have no collateral. For these to be risk free, it would be necessary that loans are sufficiently small so that there is no realized default.

To assess whether lending is risk free in the spirit of these models, we are interested in whether two firms pay the same interest rate if they issue identical loans at the same time. In other words, can the dispersion in interest rates in Figure 2(a) be accounted for by loans having different characteristics such as the time at which they were issued, the maturity, the amount, the interest rate structure (e.g. fixed versus variable), et cetera? To implement this, we use the following regression:

$$r_{lt} = \beta \mathbf{L}_l + \mathcal{M}_{lt} + \varepsilon_{lt}, \quad (1)$$

where r_{lt} is the interest rate on loan l at its initiation date t , $\mathcal{M}_{lt}$ contains maturity-time fixed effects and $\mathbf{L}_l$ is a vector of other loan characteristics.²² Throughout the paper bold capital letters will be used to denote vectors of observable control variables, and calligraphic letters will denote vectors of fixed effects. For the maturity-time fixed effects, the time dimension is monthly and loans are grouped into maturities with the following durations (in years): (0, 0.25], (0.25, 0.5], (0.5, 1], and then in yearly intervals from (1, 2] up to (29, 30]. In this way, the maturity-time fixed effects control for differences in the risk free rate across the yield curve, and any changes in the curve at a monthly frequency. From here on, we will refer to the controls in $\mathbf{L}_l$ and $\mathcal{M}_{lt}$ as “loan characteristics.”

With the loan variables in $\mathbf{L}_l$, we aim to control for all loan characteristics that are likely to affect interest rates and are available in the data: log of the loan value, frequency at which the interest rate can be adjusted, type of collateral, number of collateral assets,

²²We focus on interest rates as the outcome variable instead of spreads, since we control for maturity-time fixed effects. These fixed effects absorb variation in the risk free rate both across time and across maturities, allowing us to isolate differences in lending rates besides shifts in the underlying term structure of interest rates. See the Appendix for additional details.

amortization structure, purpose of the loan, frequency of repayment, whether the bank has the right to demand immediate repayment, whether the loan is securitized, and other special characteristics. All controls are categorical, except for the log of the loan value. For the frequency of interest rate adjustment, rates can be fixed or adjusted at a daily, monthly, quarterly, biannual or annual frequency, or at the lender's discretion.²³ For the type of collateral, the categories are no collateral, guarantees, financial assets, real estate, other collateral of a real nature, and others. For collateralized loans, the number of collateral assets can be one, two, three, four, or at least five.²⁴ For loans with more than one collateral asset, the collateral type is determined by the highest value asset. The amortization methods are French (constant installments with interest paid at the end of each period), German (constant installments with interest paid in advance of each period), fixed principal repayments, all principal repaid at maturity, and installments that increase over time, with residual cases grouped together. The purpose of the loan can be car purchases, liquidity management, exports and imports, equipment and other goods, real estate or land, or no specific purpose. The frequency of repayment is categorized as monthly, quarterly, biannual, or annual, or as a single payment of principal, a single payment of principal and interest, or as a residual case. The controls for the right to immediate repayment and the securitization of the loan are dummy variables for these characteristics. For special characteristics of the loan we have controls for whether loans were granted under Covid-related policies (including guarantees), are covered by consumer credit law, or have any other special characteristics.²⁵ For more detailed information on all controls, see the Appendix.

In taking equation (1) to the data, we split the data into loans that are fully collateralized and all others, which are mostly uncollateralized. This generates two subsamples that fit with the two theoretical approaches to risk free lending in the literature, which can be assessed separately. Figure 3 presents the results, comparing the distributions of inter-

²³There are three additional frequencies of adjustment in the data, which we also include fixed effects for. They are other frequencies less than a year, frequencies greater than one year, and a category for all residual cases. These are categories are uncommon, representing 0.05%, 0.03% and 7.75% of the sample, respectively.

²⁴We use the term "assets" broadly here, to refer to all collateral types.

²⁵The other special characteristics are: syndicated loans, loans related to real estate, and loans involved in a default process. We group these together as they only account for 0.3% of the sample.

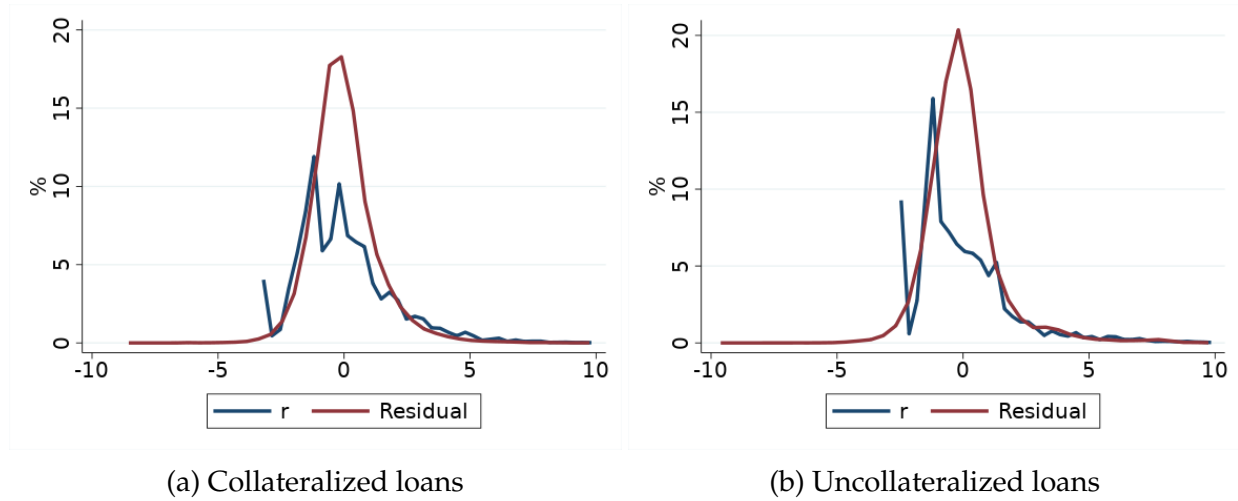


Figure 3: Interest rate distributions before and after controlling for loan characteristics. Panels (a) and (b) present the distribution of interest rates for fully collateralized and uncollateralized loans, respectively. In both panels the blue line is the unconditional distribution, and the red line is the residuals after controlling for loan characteristics. All distributions are centered with their means at zero.

est rates with the distribution of residuals from equation (1) for each sample. The means of the two distributions are aligned at zero to make them visually comparable. This comparison tells us how much variation in interest rates remains after controlling for all of the loan, bank, maturity and time characteristics that we have discussed. If firms were lending at common risk free rates, then the distribution of residuals would be a mass point at zero.

For both collateralized and uncollateralized loans, we see that substantial variation in interest rates remains after controlling for loan characteristics, showing that firms are not lending at common risk free rates. Quantitatively, 52% of the variance in interest rates remains for collateralized loans, and for uncollateralized loans it is 60%. It is possible that some types of collateral could be more effective at insuring banks against risk than others, and that some of the residual interest rate variation is due to this. To assess this, in the Appendix we repeat the analysis separately for different classes of collateral: guarantees, real and financial assets, and other forms of collateral. The results show that the type of collateral is not materially important for the conclusion. More than 41% of the variation in rates remains unexplained in all cases.

3.2 Lending with risk

With the data departing substantially from risk free lending, we now ask whether the dominant theory of interest dispersion in macroeconomic models, default risk, can account for the additional variation.

In models adopting this theory, financial intermediaries earn the risk free rate on loans, plus a premium to compensate for the risk of default, and the loss when default occurs. Financial markets are assumed to be perfectly competitive and the spread is pinned down by intermediaries earning zero expected profit. While there is variation from paper to paper in the details of the assumptions that determine default probabilities and losses given default, the vast majority of papers use single period debt and the price of loan l for firm i at time t is determined by an equation with the form

$$Q_{lt} = \mathbb{E}_t \left[\Lambda_{t+1} \left(1 - \chi(s^{l,t+1}) LGD(s^{l,t+1}) \right) \right]. \quad (2)$$

Λ_{t+1} is the stochastic discount factor, $s^{l,t+1}$ is the history of states for the firm with loan l up to time $t + 1$, χ is an indicator function for whether default occurs in a given state, and LGD (*loss given default*) is a function mapping the state for loan l to the share of the loan that is not recovered in the case of default. The expectation is taken over the set of possible states at time $t + 1$, given the state in period t . For examples of this type of setup, see Cooley & Quadrini (2001), Khan et al. (2021) and Ottonello & Winberry (2020).²⁶

Our objective is to take this equation to the data to evaluate whether default risk can explain the variation in interest rates that we observe. For our baseline approach, we follow the most common assumption in the theory, that banks are risk neutral. Later, we will investigate alternatives. Under this assumption, equation (2) can be expressed in terms of interest rates as

$$1 + r_{lt} = \frac{1 + \tilde{r}_t}{(1 - PD_{lt} LGD_{lt})}, \quad (3)$$

²⁶There are various microfoundations for the losses that arise from default. For example, Ottonello & Winberry (2020) assumes that the lender cannot recover the full value of the collateral. Kermani & Ma (2023) provided empirical evidence supporting this approach. In Cooley & Quadrini (2001) there is asymmetric information and banks incur monitoring costs in the process of enforcing their right to collateral when default occurs.

where $\tilde{r}_t$ is the one period risk free rate at time t . PD_{lt} is the overall probability that loan l issued at t defaults in the next period, and LGD_{lt} is a weighted average of loss given default across states next period, with the weights equal to share of total default risk for the loan in each state.²⁷

In order to take this equation to the data we need measures of PDs and LGDs that are consistent with the theory.²⁸ The data provides two such measures of PDs. First, banks report their estimates of the PDs of firms each month in the credit registry. Second, the banking regulator estimates these PDs each year as well.²⁹ The horizon at which the PDs are estimated is one year, and they are estimated at the firm rather than loan level, because this is the regulatory requirement. A firm is considered in default if, for three consecutive months, 2.5% or more of its total outstanding loans remain overdue. A loan is classified as overdue when there is an outstanding payment past its due date. This definition and the fact that default is measured at the firm rather than the loan level are fine from the perspective of the theory, as long as LGDs are estimated conditional on the same definition of default. Fortunately, this is precisely the case. For LGDs we have values reported by banks at the time that a loan is initiated.

From the perspective of the theory, the important thing for the PD and LGD estimates are that they are the banks' beliefs about these variables at the time they are issued, even if they are incorrect. Nevertheless, we verify the quality of the PD estimates by tracking all loans for one year after initiation, and checking that ex post default rates are in line with the ex ante estimates. There is some underestimation of PDs by both banks and the regulator, but for most of the sample default rates are approximately increasing one for one with PD values. See the Appendix for full details. For LGD values, we evaluate the potential effect of measurement error in robustness exercises below.

Since PDs are estimated at a one year horizon, we restrict the sample for the baseline analysis to loans with maturities up to one year to maintain a tight connection between the model and the data.³⁰ The PD and LGD values from banks are only available for loans

²⁷ Additional mathematical details are provided in the Appendix.

²⁸ Note that PD and LGD will be used as abbreviations for default probability and loss given default, and *PD* and *LGD* (italicized) are the variables measuring these in regressions.

²⁹ The methodology for this is explained in Antunes et al. (2016).

³⁰ To perfectly map the theory to the data, we would want to use only loans with a maturity of exactly

issued from 2018, and do not have full coverage, whereas the regulator PDs are available for the full sample. We therefore use the regulator PDs for the baseline analysis, with the banks' LGDs, and in the Appendix we show that the results also hold for the bank PDs. Overall, the baseline sample size for this analysis is 89,896 loans.³¹ In a robustness exercise discussed below, we check that the results are not driven by any sample selection.

In taking equation (3) to the data, we also want to control for the effect of loan characteristics on rates, so that we can assess the total share of the variation that can be accounted for by default risk and loan characteristics. To do this, we take logs of both sides of the equation, rearrange, and then add the controls from before to the right hand side to give the regression equation:

$$r_{lt} = \beta_1 \log(1 - PD_{lt}LGD_{lt}) + \beta_2 \mathbf{L}_l + \mathcal{M}_{lt} + \varepsilon_{lt}. \quad (4)$$

Note that we allow the coefficient on the default risk term to be determined by the data to enable the best possible fit. Since the risk free rate is purely a function of maturity and time, it is absorbed by the maturity-time fixed effects and does not need to be included in the regression. For estimating this equation, we no longer divide the sample by whether loans are collateralized or not since this is not an important dimension of the data for rate heterogeneity.

Figure 4(a) compares the actual interest rate distribution in blue, and the distribution after removing the effects of default risk and loan characteristics in red. The two distributions overlay each other, showing that risk is not a major factor in explaining the interest rate distribution. If risk accounted for the residual variation after the previous set of controls, then the red distribution would be concentrated on zero.

To quantify the amount of variation in interest rates that can be attributed to risk, we adopt the Abowd et al. (1999) (AKM) method. For each observation, let $\hat{r}_{lt}^c$ denote the estimated value of component c from the right-hand side of equation (4). The three right-hand side components are the risk term, loan characteristics ($\beta_2 \mathbf{L}_l + \mathcal{M}_{lt}$) and the

one year, because this is the horizon for the PD definition. The issue with this is that the sample size is small. We have therefore extended the sample to all loans with a maturity up to one year. This will result in an overestimate of price of default risk for these loans.

³¹The sample is distributed evenly over time for the period covered by the default risk variables (September 2018 onwards).

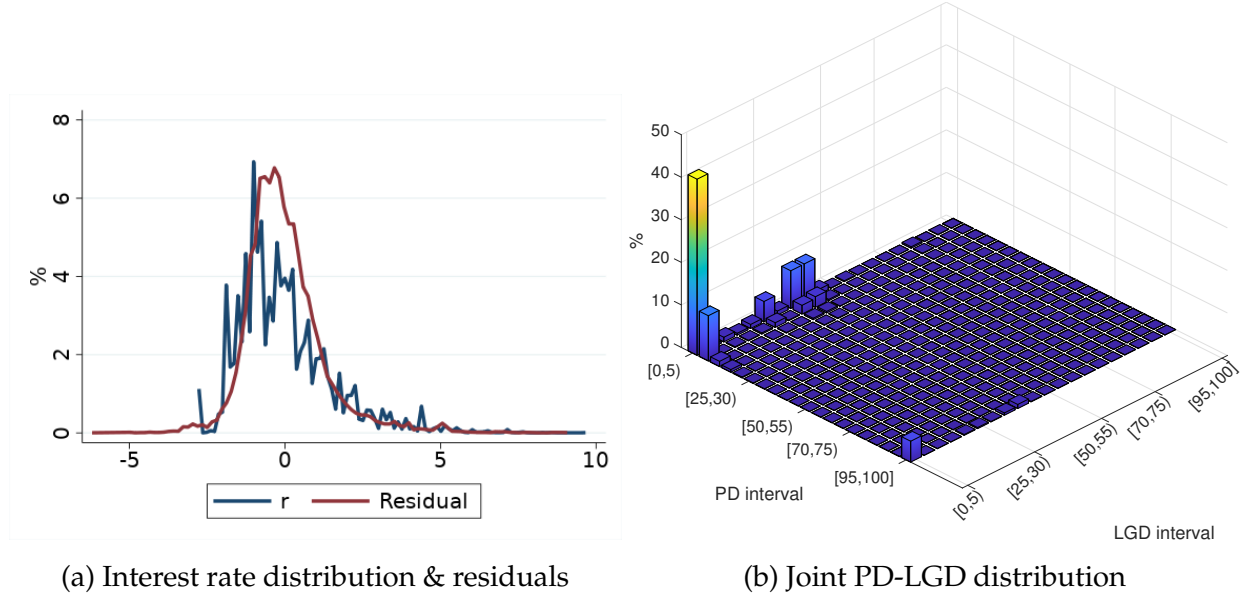


Figure 4: Interest rates and risk. Panel (a) presents the raw interest rate distribution (blue) and the residuals after controlling for loan characteristics and default risk (red). Both distributions are centered with their means at zero. Panel (b) presents the joint distribution of PDs and LGDs. PDs and LGDs are grouped into twenty bins: $[0\%, 5\%)$, $[5\%, 10\%)$, $\dots$, $[95\%, 100\%)$.

residual. The overall variance of interest rates is given by sum of the variances of each of these components plus the covariances between them:

$$\text{Var}(r_{lt}) = \sum_{c=1}^N \text{Var}(\hat{r}_{lt}^c) + \sum_{j=1}^N \sum_{c=1}^N \text{Cov}(\hat{r}_{lt}^c, \hat{r}_{lt}^j), \quad j \neq c, \quad (5)$$

where N is the number of right hand side components in the regression. The contribution of component c to the total variance is $\text{Var}(\hat{r}_{lt}^c) / \text{Var}(r_{lt})$. The results imply that default risk accounts for only 1.2% of the variance in rates and 70% of the variance is unexplained.³²

Risk is not very important for interest rates because there is simply not much risk in the data. A large fraction of firms have a PD close to zero. The median PD is 2.2% and the 90th percentile of the distribution is only 11%. The LGD distribution is equally concentrated around low values with its median being 0%. The joint distribution of these variables in Figure 4(b) highlights the low level of risk. There are some LGD values in the range of 30–50%, but the vast majority of PD and LGD value are less than 10%, and most are less than 5%.

³²Loan characteristics account for 27% of the variance and the remaining 1% is due to the covariance terms.

We have performed a wide range of exercises to verify that default risk accounts for only a small share of the variation in interest rates. We now discuss these exercises, with the full details provided in the Appendix. For exercises that involve variations to the specification in equation (4), or estimating it on alternative samples, the results are summarized in Table 2.

One potential concern is that there could be measurement error in the PDs and LGDs. We address this in three ways. First, we repeat the analysis using the PD estimates from banks. Assuming that these are accurate reports of the banks' beliefs at the time when loans were initiated, they are the correct measure of risk for pricing loans, even if they are inaccurate *ex post*. When we use these measures the results are virtually the same.³³ Second, we consider the possibility that the reported PDs do not fully reflect the information of banks. If banks have better information and price loans accordingly, then the interest rate residuals in the regression would have predictive power for defaults. We have tested this by regressing a default indicator on these residuals and find that this is not the case. The R^2 for this analysis is 0.1%.³⁴ Third, we evaluate how large measurement error would need to be in order to explain the variation in rates that is observed. Specifically, we take the estimated coefficients for equation (4) and ask what values the PDs would need to take to fit the interest rate observations holding LGDs at their values in the data and, alternatively, what values LGDs would need to take holding PDs to their empirical values. The values are implausibly large. More than 20% of PDs and LGDs would need to be higher than 100%, with the median value of both distributions more than one order of magnitude larger than the reported values.

We also evaluate the effects of alternative stochastic discount factors for pricing loans. Under the baseline approach, we are assuming that lenders are risk neutral and therefore discount at the risk free rate. In principle, lenders could discount the future at a different rate, or have a stochastic discount factor that varies across states so that if two loans have payments with different correlations with the stochastic discount factor, then they will

³³In the Appendix we show that the PDs from banks and the regulator are very similar, which explains this result. Again, most loans have low PDs and low LGDs.

³⁴For this analysis we use a more flexible semi-parametric specification for the relationship between the risk variables and interest rates when extracting residuals, to allow the regression to capture the full information content of these variables. This specification is described below.

be priced differently. For the first of these variations, note that when we take the pricing equation to the data, the stochastic discount factor is absorbed into the maturity-time fixed effects because its only variation is with respect to time. Our methodology is therefore robust to any common discount rate of lenders. To evaluate the second possibility, we care about the correlation between the likelihood that a firm defaults and the lenders valuation of the states in which default occurs. For a variable with information about the former, we choose the revenue of firms. For the value of states, we take guidance from consumption-based asset pricing theories and use the value of aggregate consumption. We take the correlation between these two variables and add it as a control to regression (4). We also allow it to interact with the risk term in this equation. This variation hardly affects the baseline results, with default risk now accounting for 1.2% of interest rate variance.

The fact that changes to the stochastic discount factor do not affect the results should not be surprising, given the evidence already presented. The loans do not have a lot of risk to begin with, implying that their payoffs are similar across states. So changing the stochastic discount factor, which affects the weighting of states, is unlikely to affect the relative prices of loans.

Equation (3) places a particular structure on the relationship between loan prices and default risk. While this is the dominant approach in macroeconomic models, it is possible that it is misspecified and that the true relationship takes another form. In the baseline regression, equation (4), we give the relationships additional freedom by allowing the coefficient on the risk term to be estimated rather than imposing a value of minus one. The estimated coefficient validates our approach, as it is -0.08 and statistically different from minus one. If we impose a coefficient of minus one, the explanatory power of default risk for interest rates decreases. To be more flexible with the relationship between interest rates and risk, we also use a semi-parametric specification. For this, we group PDs and LGDs into deciles, then take equation (4) and replace the risk term with six new terms: PD_{it} and PD_{it}^2 , allowing the coefficients to differ for each decile of PDs; LGD_{it} and LGD_{it}^2 , allowing the coefficient to differ for each decile of LGDs; $PD_{it} \times LGD_{it}$ and $PD_{it}^2 \times LGD_{it}^2$,

allowing the coefficient to vary for each possible pair of PD and LGD deciles.³⁵ This specification increases the share of rate variation due to default risk to 14.9%, indicating that the empirical relationship between default risk and interest rates deviates from the theory. Nevertheless, the majority of interest rate variance remains unaccounted for.

An alternative approach to the analysis that circumvents concerns about how risk is specified in the regression, as well as reducing concerns about PD and LGD measurement, is to restrict the sample to loans with a low PD .³⁶ If variation in rates is primarily due to loan characteristics and default risk, then for low risk subsamples, loan characteristics should account for most of the variation in rates. To test this, we restrict the sample to loans with a PD below 0.29%, 0.55% and 1.14% (the 5th, 10th and 25th percentiles of the PD distribution, respectively), and reestimate equation (1). This changes the results little. For these subsamples 53%, 54% and 57% of the variance in rates remains unaccounted for by the controls.

To ensure that our findings are not specific to the subsample of loans for which LGD values are available, we replicate the semi-parametric analysis using only PD s, as these are available from the regulator for all loans in the data. We do this in two steps. First, we hold the sample fixed and reestimate the semi-parametric specification excluding LGD s, to verify that their omission does not materially affect the results. Second, we extend the exercise to the full loan sample. Results are robust across both exercises, indicating that our findings are not driven by the particular subsample of loans with LGD data.

A limitation of the analysis is that the theory is based on a loan for which all principal and interest are repaid at maturity. This is not the case for all loans in the sample. We address this issue in two ways. First, we restrict the sample to loans for which all principal and interest are paid at maturity, providing a closer correspondence to the theoretical framework. We also reestimate the results separately for loans with French amortization and for loans with other amortization structures. The results remain quantitatively similar, with the explanatory power of default risk for interest rate variation ranging from

³⁵To allow the coefficients to vary as described we multiply each term by the relevant dummy variables for PD and LGD deciles.

³⁶Based on the theory, we would ideally like to focus on loans with the lowest values of $PD \times LGD$. We use just the PD instead, as this allows for a much larger sample.

	Risk explanatory power
<i>Baseline</i>	1.2%
<i>Robustness test</i>	
Bank PDs	2.7%
Alternative stochastic discount factor	1.2%
Semi-parametric specification	14.9%
Semi-parametric using only PDs (baseline and full samples)	5.4%, 2.3%
Amortization types (all principal at end, French, other)	2.0%, 0.4%, 0.3%
Value of exposure	0.0%
Exclusion of Covid period	0.4%
Firm size groups (from small to large)	0.2%, 0.5%, 1.0%, 1.5%, 2.1%
Firm age groups (0–4, 5–15, more than 15 years)	0.7%, 0.6%, 1.2%
All credit instruments	0.7%
Extension to longer maturity loans	0.7%

Table 2: Robustness tests for default risk results. “Semi-parametric using only PDs” refers to the semi-parametric specification in which PD values are used, but no LGD values. This analysis is done for the full sample, and the sample for which LGD values are available (i.e. the baseline sample). The amortization types are all principal paid at maturity, French amortization, and all other categories grouped together. In the exercise adjusting for the value of exposure, we multiply each LGD value by the expected exposure at default as a share of the loan value. The analysis excluding the Covid period uses loans issued before 2020 only. The five firm size groups are small (0–4 employees), medium-small (5–12 employees), medium (13–39 employees), medium-large (40–119 employees) and large (120 employees or more). The analysis for amortization types and firm size and age groups involves repeating the analysis for each of the specified subsamples.

0.3% to 2.0% for these subsamples.

Second, we directly assess whether payments on loans prior to maturity affect the results by accounting for a bank’s expected exposure at the time of default.³⁷ To incorporate this, we multiply LGD by exposure (as a share of initial loan value) to allow for repayments before default to reduce a bank’s potential losses. In principle, heterogeneity in exposure across loans could generate additional heterogeneity in rates. The analysis shows that this does not affect the main results.

We also verify that the results are not driven by the Covid period and potential guarantees offered by the government that would eliminate default risk for banks. To do this we restrict the sample to loans initiated before 2020 and show that the results are similar to our baseline. To guarantee that results are not driven by a specific group of firms, we verify that they hold across firm age and size. The results are also robust to including all

³⁷Exposure is the total outstanding amount on a loan (principal and interest) at a given point in time.

types of credit instruments in the sample. When all such instruments with a maturity of up to one year are used, instead of just firm financing loans, the sample size increases to 108,847 observations. The results are essentially unchanged for this broader sample, with risk accounting for 0.4% of rate variation.

The final exercise extends the analysis to longer maturity loans. The baseline analysis restricts the sample to loans with a maturity of up to one year because that is the horizon for default probabilities. For loans with a maturity of $T \geq 1$ years, and all principal and interest paid at maturity, equation (3) can be generalized to

$$1 + r_{ltT} = \frac{1 + \tilde{r}_{tT}}{(1 - PD_{ltT} \times LGD_{ltT})^{\frac{1}{T}}}. \quad (6)$$

r_{ltT} is the per annum interest rate for loan l issued at time t with a maturity of T years, and $\tilde{r}_{tT}$ is the per annum risk free rate at time t for this maturity. PD_{ltT} is the overall probability of default over the T year horizon, and LGD_{ltT} is a weighted average of loss given default across states in T years, with the weights equal to the share of total default risk for the loan in each state.³⁸ As for the baseline analysis, to take this equation to the data we take logs of both sides and add controls to the right hand side for loan characteristics. To extend PD estimates beyond one year, we assume that they are constant over the maturity of a loan, and the realization of default is i.i.d. each year.³⁹ LGD in the data is the share of the loan value that is expected to be lost if there is default within the next year. We assume that this value is the same for a default occurring at any point in the future for longer maturity loans. With these assumptions, we can extend the analysis to loans of all maturities.⁴⁰ We find that the results are very similar to our baseline: the variation in interest rates explained by risk is 0.7%.

Our overall finding is that default risk has small explanatory power for the variance in interest rates. At best, under the semi-parametric approach, it accounts for 14.9% of the variance. After accounting for both default risk and loan characteristics, 62% of the total

³⁸See the Appendix for additional details for equation (6).

³⁹This means that the default probability for loan l , initiated at time t , with a maturity of T years is $1 - (1 - PD_{lt})^T$.

⁴⁰For loans with some principal or interest paid prior to maturity, or an adjustable rate, equation 6 should be treated as an approximation.

variance in rates is unaccounted for.

4 Banks and relationship banking

This section turns to leading alternative explanations for interest rate dispersion. The finance literature emphasizes the role of bank heterogeneity arising from differences in cost structures, risk profiles, and market power across lenders, and firm–bank relationships, whereby repeated interactions and private information shape loan pricing, as sources of interest rate dispersion. These ideas have been incorporated into macroeconomic theory too.⁴¹ This section quantifies the contribution of each of these sources to the observed dispersion in interest rates.

4.1 Banks and interest rate variation

Bank heterogeneity can translate into differences in the interest rates that they charge on loans in several ways. The finance literature identifies three main sources of such heterogeneity. First, market power, which directly affects lending markups, as originally formalized in the models of [Monti \(1972\)](#) and [Klein \(1971\)](#). Second, cost heterogeneity, since larger or more efficient banks typically face lower funding and monitoring costs, consistent with the intermediation and delegated monitoring frameworks of [Diamond \(1984\)](#). Third, risk composition, as banks with riskier asset portfolios demand higher spreads between lending and deposit rates to compensate for default and interest rate risk ([Ho & Saunders, 1981](#); [Angbazo, 1997](#)).⁴²

The objective of our analysis is to quantify the contribution of bank heterogeneity to interest rate variation, while being agnostic about the mechanism, by using rich fixed effects. Our baseline approach takes regression (4), and adds bank and firm fixed effects. We add both types of fixed effects to negate the risk that firms sort across banks in a way

⁴¹For example, see [Corbae & D’Erasmus \(2021\)](#), [Bellifemine et al. \(2022\)](#), and [Dempsey & Faria-e Castro \(2025\)](#).

⁴²Empirical evidence supporting these mechanisms includes [Berger & Mester \(1997\)](#); [Berger & Hannan \(1998\)](#); [Wheelock & Wilson \(2012, 2018\)](#); [Darmouni \(2020\)](#); [Benetton \(2021\)](#). Moreover, [Corbae & D’Erasmus \(2021\)](#) document systematic markup heterogeneity across banks of different sizes and develop a quantitative model that nests these three channels. More recently, [Bellifemine et al. \(2022\)](#) propose a heterogeneous-bank framework to quantify the role of banks in the propagation of macroeconomic shocks.

that would result in the bank fixed effects picking up what are actually firm effects. The regression for loan l of firm f from bank b initiated at time t is:

$$r_{lfbt} = \beta_1 \mathbf{R}_{lt} + \beta_2 \mathbf{L}_l + \mathcal{M}_{lt} + \mathcal{B}_b + \mathcal{F}_f + \varepsilon_{lfbt}, \quad (7)$$

where $\mathbf{R}_{lt}$ contains controls for default risk, $\mathcal{B}_b$ and $\mathcal{F}_f$ denote bank and firm fixed effects, respectively, and the other controls are defined in the same way as in the previous section. From now on we control for risk using the semi-parametric method introduced in Section 3.2, including only PD terms.⁴³ We omit LGD terms since including these reduces the sample size substantially, and we need a large sample in order to estimate bank and firm fixed effects. The robustness exercises in Section 3.2 showed that leaving out LGD controls decreases the explanatory power of risk by around 50%. We are interested in how much of the variation in rates can be accounted for by the bank fixed effects, $\mathcal{B}_b$. We emphasize that these fixed effects capture the effects of *all* relevant bank characteristics that are common over time and affect all firm financing loans in a bank's portfolio.

The bank and firm fixed effects in this setting are analogous to the firm and worker fixed effects, respectively, that are estimated in the labor literature on the determinants of wages. In that literature, identification of the fixed effects typically relies on variation arising from people changing jobs, which can generate mover bias.⁴⁴ Our identification does not rely on the same variation and is not subject to this issue. The banking side of the network is highly concentrated: a small number of large banks lend to many distinct firms, so bank fixed effects are estimated with large sample support and are well identified. At the same time, firms in the sample have multiple loans with different characteristics, even when they borrow from a single bank, providing additional within-firm variation that helps precisely estimate firm fixed effects conditional on well-identified bank effects. Put differently, identification of the two sets of effects is contingent on dense bank-to-many-firms links and substantial within-firm loan variation. The empirical network is

⁴³Specifically, we interact dummy variables for each decile of the PD distribution with dummy variables for loan maturities, and multiply each of these by linear and quadratic terms for the PD. This allows the coefficients on PD and PD^2 to vary for each PD decile-maturity pair. For measuring PDs we follow our baseline approach of using the value from the regulator to maximize sample size.

⁴⁴By mover bias we mean the bias that can arise when fixed effects are identified from people who move jobs and job mobility is non-random.

	Contribution
(1) Default risk	0.0049
(2) Loan characteristics	0.2555
(3) Bank fixed effects	0.0956
(4) Firm fixed effects	0.3265
(5) Residual	0.2766
(6) Covariances	0.0458
# Loans	521,856

Table 3: **Contributions of default risk, loan characteristics, and bank and firm fixed effects to interest rate variation.** Each line provides the contribution of the one of the components of the right hand side of equation (7) to overall interest rate variance. The contributions are computed using equation (5).

therefore highly connected and we find no evidence that only a small, unrepresentative subset of multi-bank firms drives our results. In the Appendix we report the distribution of the number of loans and banks per firm, and the distribution of the number of loans and firms per bank. Additionally, we later show that results do not change when restricting the sample to only firms with multiple bank relationships.

Table 3 reports the AKM decomposition of the variance in interest rates, presenting the contribution of each component of regression (7). The main result is in the third line: banks only account for 9.6% of the variance in interest rates and are therefore not a primary source of dispersion. Loan characteristics (which include maturity-time effects) remain important, accounting for 25.6% of variation, and the other standout contribution comes from firm fixed effects, which account for 32.7%. We will further evaluate firm effects in Section 5. An additional point to note is that the covariances between the components of the regression matter little for overall interest rate variance.⁴⁵

As an extension of the analysis, we also consider a specification in which the bank and firm fixed effects, $\mathcal{B}_b$ and $\mathcal{F}_f$ respectively in equation (7), are replaced by bank-time and firm-time fixed effects. This specification allows the effects of individual banks on loans to vary over time. In order to estimate this specification, we need observations for firms that

⁴⁵Neither the sum of the covariances nor the individual terms contribute substantially to the overall variance. For example, the largest covariance term—between bank fixed effects and loan characteristics—accounts for 6% of the total variance. The relatively low importance of the covariance terms indicate that there is little systematic sorting of specific firms (banks) into particular banks (firms) or loan characteristics.

originate multiple loans in the same period. There is a tradeoff between the sample size and frequency of variation that is permitted.⁴⁶ We use a one year time period, judging that this is sufficiently high frequency to capture changes in firm and bank characteristics that could affect interest rates, while preserving the sample size. This gives us a sample of 41,312 firms, relative to a sample of 84,535 under the baseline approach. The results are presented in the Appendix and show that allowing bank effects to vary with time does not change the contribution of banks to rate variation: under this specification it is 9.5%.

For additional verification of this result, we perform robustness exercises. One potential concern is that the firm fixed effects might be upwardly biased, and impact the bank fixed effect estimates, if there are not enough loans per firm. In the baseline analysis there are six loans per firm on average. We perform two exercises to address this. First, we restrict the sample to firm with at least 10 loans each and, second, we use all instruments instead of just firm financing loans. These modifications increase the average number of loans per firm to 36 and 12, respectively. For both specifications the results are similar to the baseline. Bank fixed effects account for less than 10% of rate variation, and firm fixed effects account for 42% and 37%, respectively.

Similar to our analysis in Section 3.2, we test the results by amortization type, by firm size and age groups, and exclude the Covid period. Overall, results are robust across all of these subsamples, with bank effects never explaining more than 15% of the variation in rates. Full details of these exercises are in the Appendix.

4.2 Firm-bank relationships and interest rate variation

While bank heterogeneity does not play a large quantitative role in accounting for interest rate variation, there could be additional important effects at the level of individual bank-firm relationships. Relationship banking theory holds that repeated interactions between firms and banks can affect credit terms. One rationale for this is informational: by learning about a borrower over time, banks can overcome adverse selection and moral hazard problems (e.g. Diamond, 1984; Ramakrishnan & Thakor, 1984; Von Thadden, 1995). A

⁴⁶As the frequency of the time dimension of the fixed effects increases, the number of firms that originate multiple loans a single period falls, decreasing the sample size.

second rationale is contractual: long-term relationships enable intertemporal flexibility and implicit risk sharing through rate smoothing and covenant adjustment, as modeled by Boot et al. (1993), Boot & Thakor (1994), and Berlin & Mester (1992, 1998).⁴⁷ Relationships can also affect loan prices by generating monopoly power for banks (e.g. Sharpe, 1990; Rajan, 1992) and leading banks to refinance weak firms (e.g. Bolton & Scharfstein, 1996; Faria-e-Castro et al., 2024b).

The objective of this section is to quantify the relevance of factors that are specific to individual relationships between firms and banks for overall interest rate variation, while being agnostic about the specific mechanisms. We again take a fixed effects approach. In doing so, it is important to separate factors that pertain to the relationship itself from those that are idiosyncratic to each firm or bank. The analysis makes use of regression (7), as well as the following variation of it:

$$r_{lfbt} = \beta_1 \mathbf{R}_{lt} + \beta_2 \mathbf{L}_l + \mathcal{M}_{lt} + \mathcal{FB}_{fb} + \varepsilon_{lfbt}, \quad (8)$$

where $\mathcal{FB}_{fb}$ is a vector of firm-bank fixed effects and the other variables have the same definitions as they have had throughout the paper. The difference with regression (7) is that the firm and bank fixed effects are now interacted. In order to isolate the effect of the relationships between firms and banks on interest rates, there are three steps. First, we estimate regression (8) and compute the contribution of firm-bank fixed effects to total interest rate variation using the AKM method. This provides a measure of the total contribution of firm factors, bank factors and relationship factors jointly.⁴⁸ Second, we take the same sample of loans, reestimate equation (7), and use the AKM method to measure the separate bank and firm contributions to interest rate variation. Third, the difference between the sum of these values and the total contribution of firm-bank fixed effects in step one, is the contribution of the pure relationship component. For this analysis, firms need to have multiple loans from multiple banks. This restriction results in a sample of 463,624 loans to 73,858 firms.

⁴⁷Empirically, Petersen & Rajan (1994, 1995) and Berger & Udell (1995) show that stronger, more exclusive relationships lower borrowing costs, reduce collateral requirements, and ease credit access, consistent with theories of intertemporal rent-sharing.

⁴⁸Note that the sample is smaller than for the previous analysis because firms need to have received loans from multiple banks to be included.

	Contribution
(1) Bank effects	0.095
(2) Firm effects	0.361
(3) Firm-bank relationship effects	0.115
# Loans	463,624

Table 4: Contributions of firm, bank and relationship effects to interest rate variation.

The values are computed by estimating equation (8), and then equation (7) on the same sample. For each estimation we decompose the interest rate variance using equation (5). The contributions of the bank effects and the firm effects are the estimated contributions of the relevant fixed effects from equation (7). The firm-bank relationship effects contribution is the difference between the contribution of firm-bank fixed effects from equation (8), and the total contribution of firm and bank fixed effects from equation (7).

The results are presented in Table 4. Firm-bank relationships account for 11.5% of the overall variation in rates and, once this component is removed, the contribution of pure bank effects is still 9.5%. So, in total, banks and relationship banking account for 21% of variation. While this is a meaningful amount, we are still only accounting for 53% of the variance in rates after including variation due to loan characteristics, default risk, bank effects, and relationship banking. For the remaining variation, firm fixed effects are still the standout factor, responsible for 36.1% of variation. As well as being quantitatively important, the relevant characteristics of firms are unrelated to their banking relationships. This analysis also shows that when the sample is restricted to firms with multiple bank relationships, it does not alter the importance of the firm fixed effects, reinforcing the validity of our fixed effects identification.

To provide further support for these results we perform additional exercises that we report in the Appendix. Since firm-bank relationships can change over time, our baseline estimate may underestimate their contribution. To evaluate this, we modify the regressions by adding a time dimension to the firm, bank and firm-bank fixed effects.⁴⁹ The frequency for the time dimension is one year. We find that this has a modest effect on the results, with firm-bank relationships accounting for 16.4% of the total variation in interest rates. We also extend the sample to include all credit instruments and relationship banking accounts for 17.6% of variation.

⁴⁹This means that equation (8) is estimated with firm-bank-time fixed effects and equation (7) is estimated with firm-time and bank-time fixed effects.

5 Firm characteristics and interest rates

The results in the previous section indicated that firm characteristics are important for accounting for interest rate variation across loans. In this section we investigate what characteristics are important. We evaluate a broad range of observable characteristics alongside unobservable factors, and complement this with a statistical method that determines whether differences in rates across firms originate before or after their first loans.

To understand what firm characteristics are important for determining interest rates, we take a two-step approach. In the first step we take regression (7) and add controls for relevant time-varying firm characteristics, keeping firm fixed effects to account for permanent heterogeneity in interest rates not accounted for by the observable characteristics used as regressors. We then take the estimated fixed effects and evaluate how much of these can be accounted for by observable permanent firm characteristics.⁵⁰

The regression for the first step is:

$$r_{lfbt} = \beta_1 \mathbf{R}_{lt} + \beta_2 \mathbf{L}_l + \mathcal{M}_{lt} + \mathcal{B}_b + \beta_3 \mathbf{F}_{ft} + \mathcal{F}_f + \beta_4 \mathbf{BF}_{bft} + \varepsilon_{lfbt}, \quad (9)$$

where $\mathbf{F}_{ft}$ is a vector of control variables for firm f at time t , $\mathbf{BF}_{bft}$ is a vector of controls for characteristics of the relationship between firm f and bank b at time t , and all other variables maintain their existing definitions. We are adding controls for the characteristics of firm-bank relationships rather than using the methodology from Section 4.2, which requires firm-bank fixed effects, to maintain a broad sample and economize on the number of fixed effects to estimate.⁵¹ In selecting the characteristics of firms to use in $\mathbf{F}_{ft}$, we aim to capture the state of the firm at the time the loan is granted by including everything that would be a state variable in relevant models of firms, as well as anything else in the data that could plausibly affect the interest rate charged by a bank. Note that we are already

⁵⁰The second step is necessary because the effect of time-invariant observables cannot be separately identified from firm fixed effects in the first step.

⁵¹Specifications with firm-bank fixed effects are the most demanding that we run. Substituting them with measurable characteristics of firm-bank relationships creates the risk of overestimating the role of firm characteristics and firm fixed effects. However, this does not appear to be a major concern in our setting since the estimated role of fixed firm characteristics does not change substantially when firm-bank effects are considered—see Section 4.2.

controlling for default risk, loan characteristics, bank effects, and banking relationships, so we are assessing the relevance of firm characteristics for interest rates beyond these factors.

(a) Time-varying characteristics		
Non-financial variables		
<i>Production inputs</i>	<i>Revenue & profit</i>	TFP
Capital stock	Revenue	Investment rate
Number of employees	Revenue growth rate	Inventories
Wage bill	Profit share	
Financial variables		
<i>Balance sheet</i>	<i>Credit history</i>	<i>Trade credit</i>
Equity	Past default indicator	Trade payables/total assets
Financial assets	Past late repayment indicator	Trade receivables/total assets
Total assets		
Total liabilities		
(b) Time-invariant characteristics		
Industry	Home country	Birth year indicators
Location	Legal form	Volatility
<i>Number of local banks</i>	Number of R&D employees	Std. $\Delta \log(\text{revenue})$
Top 5 banks	<i>Foreign operations</i>	Skewness $\Delta \log(\text{revenue})$
Other banks	Exporter indicator	Std. $\Delta \log(\text{TFP})$
Number of establishments	Number of foreign estabs.	Skewness $\Delta \log(\text{TFP})$

Table 5: Controls for firm characteristics. TFP is measured using the method from [Akerberg et al. \(2015\)](#). For credit history, the indicators record whether a firm has previously defaulted on a loan or been late with a repayment. A payment is classified as late if it is 30 days or more overdue. Location is measured at the 4-digit postcode level. Industries are 5-digit, implying 833 industries in total. Legal forms of businesses are private limited liability company, public limited company and other types. For the number of local banks, we count the number of top 5 banks and the number of other banks with branches within the 5-digit and 7-digit postcodes of a firm's headquarters, giving four variables in all. The number of establishments, the number of foreign establishments and the number of R&D employees are the maximum numbers observed for a firm in the data. The following variables are specified in log units in the regression: production input variables, revenue, TFP, financial assets, total assets, total liabilities, and the number of R&D employees. For inventories, there is an indicator variable for values of zero, and otherwise it is in log units. Full details for the measurement of all variables are in the Appendix.

The full list of controls for time-varying firm characteristics is provided in Table 5(a). We control for standard factors of production—total factor productivity (TFP),⁵² capital and labor (employment and the wage bill)—as they are commonly state variables for a

⁵²Measured using the method from [Akerberg et al. \(2015\)](#).

firm or otherwise are informative about production technologies. The investment rate contains information about the demand for financing, and revenue and profit are informative about the scope of the firm, its cash flow, and its ability to continue operating. Revenue growth contains information about the future trajectory of a firm. On the financial side, we control for the values of equity, financial and total assets, as well as total liabilities, to capture the general balance sheet position of firms. Credit history variables are included for any effects on loan pricing beyond default risk. The value of trade payables and receivables relative to the size of the firm provide information on cash flows. Continuous variables, and those that take a wide range of integer values, are mostly entered in the regression in log units.⁵³ For additional details about the variables and their measurement, see the notes to Table 5 and the Appendix.

To control for firm-bank relationships, we include the number of banks that a firm has borrowed from, the Herfindahl-Hirschman index for the distribution of a firm's credit across banks,⁵⁴ and five variables to capture the duration, depth and nature of the firm's relationship with the bank issuing the loan in question. For duration, we use the time since the firm received its first loan from the bank. To capture the depth and nature of the relationship we use four moments: the number of loans issued by the bank to the firm over the past five years; the log total value of these loans; and the number of loans and the value of these as a share of the number of loans and value of loans that the firm has received from all banks over the past five years.⁵⁵

In the second step of the analysis, we take the estimated firm fixed effects from regression (9), and evaluate how much of the variation in these can be accounted for by observable time-invariant firm characteristics. For this, we use the regression:

$$\mathcal{F}_f = \beta \mathbf{F}_f + \varepsilon_f, \quad (10)$$

⁵³Specifically, we use log units for capital stock, number of employees, wage bill, revenue, TFP, financial assets, total assets, and total liabilities. For inventories, there is an indicator variable for values of zero, and otherwise it is in log units. Equity is specified in levels because of negative values.

⁵⁴The number of banks is the total number of banks that a firm has obtained firm financing loans from prior to the month of loan initiation, and the Herfindahl-Hirschman index is computed using all of a firm's outstanding firm financing loans in the month prior to loan initiation.

⁵⁵For all the firm-bank relationship variables, we only consider firm financing loans.

where $\mathbf{F}_f$ is a vector of time-invariant firm characteristics. The variables included in $\mathbf{F}_f$ are summarized in Table 5(b). We control for industry to capture industry specific demand and supply factors that may not be captured by firm-level variables. For location we use controls at the 4-digit postcode level.⁵⁶ These address any location-specific demand or supply factors affecting the firm, as well as geographic variation in banking markets that could affect interest rates. To further address the potential for local banking market effects, we control for the number of top five banks and the number of other banks with branches located within the more granular 5-digit and 7-digit postcodes (four variables).⁵⁷ To capture the scope of a firm’s domestic operations, we use the maximum number of establishments that the firm attains in the data. A firm’s country of origin may influence its outcomes by changing its familiarity with the local market, the distance of firm headquarters from local operations (Giroud, 2013), or the ease with which banks can enforce contracts. The legal form of businesses has been shown to correlate with firm outcomes, perhaps as an indicator of the owner’s ambitions for the firm (Levine & Rubinstein, 2017). We control for R&D intensity with the maximum number of R&D employees that we observe for each firm.⁵⁸ Controls for whether a firm has ever exported and its number of foreign establishments address effects of international trade and multinational production.⁵⁹ Indicator variables for birth year address cohort effects, such as the interest rate level when the firm was born. Finally, we want controls for the distributions of supply and demand shocks that firms face. We proxy for these with the standard deviation and skewness of the observed path of TFP and revenue for each firm.⁶⁰

⁵⁶These correspond to 684 geographic units, with an average area of 135 square kilometers (52 square miles). For the Lisbon metro area, there are 170 such codes, with an average area of 18 square kilometers (7 square miles).

⁵⁷Five-digit postcodes correspond to 3,851 geographic units, with an average area of 24 square kilometers (9 square miles). For the Lisbon metro area, there are 764 such codes, with an average area of 3.9 square kilometers (1.5 square miles). Details about 7-digit postcodes are provided in Section 2.

⁵⁸In principle, the number of R&D employees is time-varying. However, it does not have much variation in the data, so we take its maximum value for each firm, and treat it as a fixed firm characteristic.

⁵⁹Some of the variables that we use for constructing time-invariant firm controls have a small amount of variation within firms in the data. This applies to the number of local banks, the number of establishments, whether the firm has R&D employees, whether a firm exports, and its number of foreign establishments. However, since the amount of variation within firm for these variables is very small, we measure them in ways that make them time-invariant. Otherwise they would partially capture firm fixed effects in the first stage, and make the separation between time-varying and fixed firm characteristics less clear.

⁶⁰Overall the sample has 109,657 fewer loans and 25,503 fewer firms than in the Section 4.1 analysis, as

	Contribution
(1) Default risk	0.0040
(2) Loan characteristics	0.2240
(3) Bank fixed effects	0.0931
(4) Time-varying firm characteristics	0.0093
(5) Relationship banking controls	0.0010
(6) Firm fixed effects	0.3050
(6a) Fixed firm characteristics	0.039
(6b) Latent firm fixed effects	0.267
(7) Residual	0.2871
(8) Covariances	0.0904
# Loans	412,199

Table 6: **Variance decomposition including firm controls.** Each row is the share of the total variance in interest rates due to each component of regression (9), calculated with equation (5).

Table 6 presents the contribution of each component of the right hand sides of the first and second stage regressions to the overall variance of interest rates. We are particularly focused on the contributions of firm characteristics and the latent firm fixed effects in rows (4), (6a) and (6b). The first result is that firm fixed effects explain 31% of the total variance, in line with the results from Section 4. This means that the addition of the time-varying firm observables does not change the quantitative importance of the firm fixed effect. Second, nearly all of the firm fixed effects are due to latent factors. Observable characteristics account for 3.9 percentage points of these fixed effects, with the remaining 26.7 percentage points being latent. Overall the strong firm fixed component in interest rates has a low correlation with both time-varying and invariant firm characteristics.

Robustness exercises, reported in the Appendix, extend the analysis to the sample of all credit instruments. Using all instruments results in an estimate for the contribution of firm fixed effects to total interest rate variation of 36%, with time-invariant firm observables only accounting for 3 percentage points of this.

We conduct an additional exercise to assess whether the persistent heterogeneity in borrowing costs is already present when firms first enter the credit market, or instead

not all additional control variables are reported by all firms, and loans issued to age zero firms are dropped because we cannot measure control variables that require lagged observations (e.g. growth and investment rates). The sample is now 412,199 loans and 59,032 firms.

emerges through shocks realized later in their borrowing histories. To do this, we apply the statistical framework of Sterk et al. (2021) to decompose firm-level sequences of interest rates on loans into an ex ante component and an ex post component.⁶¹ We find that the ex ante component is quantitatively important, indicating that heterogeneity present prior to first loan issuance is a significant source of persistent rate dispersion. Quantitatively, ex ante factors account for roughly 15 percentage points of total interest rate variance, compared with 31 percentage points attributed to the firm fixed effect in Table 6. Further details are provided in the Appendix.

6 Interpretation of firm fixed effects

The analysis has shown that a substantial share of the variation in interest rates across firms is due to latent, highly persistent firm-level heterogeneity. This accounts for 27% of the total variation in interest rates, and is the single largest source of this variation. To quantify this in a different way, the standard deviation in rates due to this persistent heterogeneity is 60 basis points.⁶² In this final part of the paper, we discuss potential sources of this variation. Understanding the source is an important step towards pinpointing the underlying frictions in financial markets, and evaluating their broader economic implications.

Constraints imposed by the empirics Our starting point is to specify the characteristics that an explanation must have to be consistent with the empirical analysis. There are three necessary properties. First, the explanation must be at the firm level. It needs to account for idiosyncratic variation in interest rates across firms, and it cannot be a result of differences in default risk, differences in the types of loans that firms issue, selection of firms across banks that charge different rates, or firms attaining idiosyncratic rates because of individual relationships with banks. These possibilities are ruled out by our controls for default risk, loan characteristics, bank and bank-time fixed effects, and our analysis of re-

⁶¹Sterk et al. (2021) apply the methodology to employment dynamics. Here we adapt it to interest rates.

⁶²This is the share of rate variation due to latent firm fixed effects multiplied by the total standard deviation of rates in our sample.

lationship banking.⁶³ Thus the focus should be on factors that are idiosyncratic to firms.

Second, it must be able to explain highly persistent differences in interest rates across firms. The empirical analysis says that it is firm *fixed effects* that are the main source of variation. If the source of variation was idiosyncratic to firms but time-varying, then the fixed effects estimates would be smaller, and more variation would be captured by the residuals of the regressions. Explanations that generate transitory differences in rates across firms are therefore not enough. The explanation needs to be based on factors that are idiosyncratic to firms and highly persistent or permanent.

Third, the explanation must have a low correlation with the broad range of observable characteristics that we have controlled for. Without recapping the breadth of these controls—discussed in the previous section—let us highlight that we have controlled for differences across firm size, productivity level, a range of properties of productivity processes, capital, industry and an array of measures of the financial positions of firms. Given the extent of these controls, the results indicate that the explanation for the variation in interest rates lies outside of the firm characteristics that are typically considered.

Bank-related explanations The analysis has already considered the role of bank heterogeneity and relationship banking for interest rate variation in Section 4. Here we discuss alternative channels through which banks could generate firm fixed effects in interest rates, and explain why these explanations are challenged by the data.

If banks have market power, then this could generate persistent differences in interest rates across firms. The difficulty with this, is that the controls in the analysis address many types of market power. Bank controls address the ability of banks to charge consistently different prices across their loans, and bank-time controls make this more specific, addressing time variation in differential pricing of this type. If there is geographic variation in banking markets, then rate variation could show up across local markets rather than across firms (e.g. [Rice & Strahan, 2010](#); [Bordeu et al., 2025](#)). However, the firm-level analysis included firm location controls at the 4-digit postcode level that would capture this. We have also run this analysis (results in the Appendix) with richer controls at the

⁶³For default risk, recall that we performed a number of robustness exercises to rule out there being a plausible amount of default risk that is priced by banks, but unobservable to the econometrician, that can account for the unexplained variation in rates.

5-digit postcode level and still find small effects. To address the possibility that market power shows up at the bank-location level, we also test interacting bank fixed effects with 4-digit postcode dummy variables. This does not decrease the relevance of firm fixed effects for rate variation.

Alternatively, banks could generate firm fixed effects in interest rates through pricing a collection of loans jointly, rather than pricing each loan independently (e.g. Degryse & Van Cayseele, 2000).⁶⁴ If this were to account for firm fixed effects in rates, then the effect on a firm's interest rates would need to be common across the banks that it has loans with. It would not be enough that a firm concentrates most of its loans with one bank, and gets a different price because of this. That would show up empirically as a firm-bank effect. We have already run a specification that evaluates the relevance of the scope of a firm's borrowing from a bank when we controlled for the number and value of loans in a firm-bank relationship in the previous five years in Section 5. These controls accounted for little of the variation in rates. We push this analysis further with two additional exercises. First, we take residualized interest rates after controlling for loan characteristics, default risk and bank effects, and assess whether these differ across banks where a firm has many loans and banks where it has few. We also split the sample into firms whose financing is concentrated in one bank compared to firms with very dispersed financing. In both exercises we find small differences, suggesting that the joint pricing of loans does not have a significant impact on price variation.

Another way that this channel could affect interest rates is if the owner or manager of a firm has a personal financial relationship with a bank that affects the pricing of the firm's loans. For example, a firm could get a better rate at a bank where the owner has a mortgage as well. This channel also faces the issue that the effect would need to show up across the banks that a firm borrows from, and not just at an individual bank. To assess this explanation, we run two tests in the data. We evaluate the relevance of firm fixed effects for rate variation across the size distribution of firms.⁶⁵ The assumption is

⁶⁴Degryse & Van Cayseele (2000) provides evidence that firms with more products from a bank, and greater turnover on their account, pay lower rates.

⁶⁵For this, we do two things: reestimate the results on subsamples by firm size; take the estimated fixed effects for the full sample, and compute the total variance of these within each firm size group. The second exercise looks at total variance rather than shares of variance.

that the personal finances of the owner or manager will be smaller relative to the loans of a larger firm, and therefore less relevant to the pricing of its loans. We also split the sample into public and private firms, as the personal connection between a firm and its founder or manager is weaker for public firms. The results of both exercises count against the link between personal finance and business finance being quantitatively relevant for our results: the relevance of firm fixed effects for interest rate variation hardly diminishes with firm size, and is no smaller for public firms than private firms.

Unobservable variation in demand A second class of explanations that warrants discussion is unobservable idiosyncratic shocks to firms that affect their demand for loans. For example, firms may suffer cost shocks or shocks to their investment demand (e.g. an unobservable investment opportunity) that change their willingness to pay for a loan. These shocks are firm specific, and could be uncorrelated with firm observables. However, it is not enough for such shocks to cause temporary changes in interest rates paid by firms. The second requirement is that they generate highly persistent differences in rates. So, it would be necessary that some firms persistently have different investment demand, or have a persistently different liquidity position so that they routinely have more urgent financing needs. However, if one firm has persistently different investment demand to another, then over time this will result in a different capital stock, which is controlled for. If the source of variation is persistent differences in liquidity needs, then it is hard to square this with the low correlation between the financial positions of firms and their interest rates. In addition, we estimate credit demand and supply shocks, following [Amiti & Weinstein \(2018\)](#). We then average the demand shocks over time within firms and include them as a control in regression (10) to see if they have explanatory power for the firm fixed effects. Results in the Appendix show that average demand shocks do not help to explain these effects.

Search frictions Considering the institutional setting, search frictions are a natural consideration as a source of interest rate variation. The typical process for a firm to obtain a loan is that it applies to a bank, providing a range of information, the bank processes the application, and then the firm may receive a loan offer. Firms can go through this process

at multiple banks, but it is costly.⁶⁶ Survey and empirical evidence support the existence of search frictions in this setting (Federal Reserve Bank of New York, 2014; Mazet-Sonilhac, 2025), and it is well understood from search theory that these can result in price dispersion (e.g. Salop & Stiglitz, 1977; Burdett & Judd, 1983).

Within the framework of a search model, we would need two ingredients to generate firm fixed effects in interest rates: a source of rate differences that is uncorrelated with contemporaneous firm observables, and a source of persistence for interest rates.

Focus on the first of these ingredients and consider an environment in which firms have different abilities for searching for loans, and that ability affects the distribution of rates that a firm draws from once it matches with a bank. Firms can differ in ability due to different efficiencies in applying for loans, differences in the information that firms have about the rates that banks offer, or differences in bargaining ability.⁶⁷ Since these differences in ability are unobservable, they will generate differences in interest rates, at least for initial loans. Conditional on ability, rates can still vary across firms if there are idiosyncratic effects within or across loan officers, or if rates are affected by non-economic factors like personal relationships between bank employees and firm owners, or the socio-demographic characteristics of the owner.⁶⁸ Thus, there are two potential sources of unobservable rate variation at the firm level in this setting.

For thinking about persistence, consider rate variation due to unobservable ability. If differences in ability are persistent, for example because ability is determined by owner characteristics, then this will be enough to generate persistent rate differences. If, instead, the source of variation is loan officer effects or non-economic reasons, then rates would not necessarily be persistent across loans or banks. However, the search setting provides natural reasons for persistence. If a firm that is lending from one bank can search “on-

⁶⁶The types of costs that we have in mind are the time and effort of the firm’s staff who are making the application, the time taken for the bank to process the application, as well as any costs incurred in paying lawyers, accountants or other bodies to prepare the relevant documents.

⁶⁷For evidence of heterogeneity in search efficiency (or cost), information, and bargaining ability in the context of consumer finance, see Allen et al. (2014), Allen et al. (2019), Agarwal et al. (2024) and Berwart et al. (2025).

⁶⁸There is an extensive literature documenting idiosyncratic variation in the decision making of loan officers. See Bertrand & Burietz (2023) for a summary of this. For particular evidence of discrimination on socio-demographic characteristics of business owners, see Brock & De Haas (2023).

the-job” (to borrow the language of the labor search literature) and switch if it finds a lower rate, then the current rate will affect future rates.⁶⁹ Another plausible source of persistence is that new banks observe rates on previous loans as part of the application process, and this affects the negotiated rate.

A challenge for search-based explanations is that the correlation between a firm’s interest rates and its observable characteristics needs to remain low over time. It is clear that the mechanisms outlined can result in rate differences that are uncorrelated with firm observables for the first loans that firms receive. However, if one firm persistently has a lower rate than another, then under most models of investment this will result in the lower rate firm accumulating a larger capital stock. The question is whether this mechanism is weak enough in a search model for bank loans, combined with a standard investment model, to quantitatively match the latent firm fixed effects in interest rates. In fact, this is a challenge for any theory of persistent interest rate differences once endogenous investment choice is considered.

In terms of implications, understanding the relative strengths of the proposed explanations is important. If there are fundamental reasons for the market for bank-firm loans to be decentralized and require search, and firms inherently have different search abilities, then the observed equilibrium could be efficient. If, instead, rate differences are a result of idiosyncratic behavior by loan officers or differentiation for non-economic reasons, then they are wedges that can cause misallocation, and a more standardized approach to rate setting could yield benefits. It is also worth reiterating that we have documented persistent, rather than transitory, differences in rates. These would generally have a greater impact on firm investment decisions, increasing the importance of these questions.

7 Conclusion

Firm financing is an increasingly important component of modern macroeconomic theory and interest rates are key prices. However, we still have a limited understanding of the

⁶⁹For on-the-job search to be relevant, firms would need to have the option of continuing to lend with their existing bank on similar terms. In support of persistent firm-bank relationships, 91.5% of loans in the sample are received from banks that the relevant firm received their previous loan from, or which the firm has another active loan from at the time of issuance.

empirical determinants of interest rates. This paper addresses this question with analysis made possible by a detailed new credit registry, matched with firm balance sheet and income statement data.

We show that interest rates vary substantially even after controlling for a rich set of loan characteristics such as maturity, issuance date, and the amount and type of collateral. Furthermore, only a small share of this variation is accounted for by the core theory of rate dispersion, default risk. Theories for dispersion based on bank heterogeneity and relationship banking also have quantitatively low explanatory power. Instead, the main source of variation is persistent idiosyncratic firm-level heterogeneity that is uncorrelated with a broad range of firm observables, including firm age, size, industry, location, financial position, credit history, productivity, production input mix and many others.

These findings suggest that there are substantial additional frictions in credit markets, outside of existing frameworks. The evidence implies that the main source of interest rate variation must satisfy three properties: it must operate at the firm level; generate highly persistent differences in rates across firms; and be only weakly correlated with the broad set of firm characteristics observed in the data, including the standard characteristics that define firms in macroeconomic models.

We evaluate several candidate explanations that could satisfy these requirements. The evidence provides little support for explanations based on local bank market power, the joint pricing of multiple loans to the same firm, interactions between entrepreneurs' personal and business finances, or unobservable demand shocks. Instead, it points most strongly toward search-related frictions in the market for bank loans to firms. An important task for future research is to distinguish among the specific channels through which these frictions operate, such as differences in firms' ability to search for credit, heterogeneity at the loan-officer level, or differentiation based on non-economic characteristics, and to quantify their implications for investment, misallocation, and the aggregate economy.

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